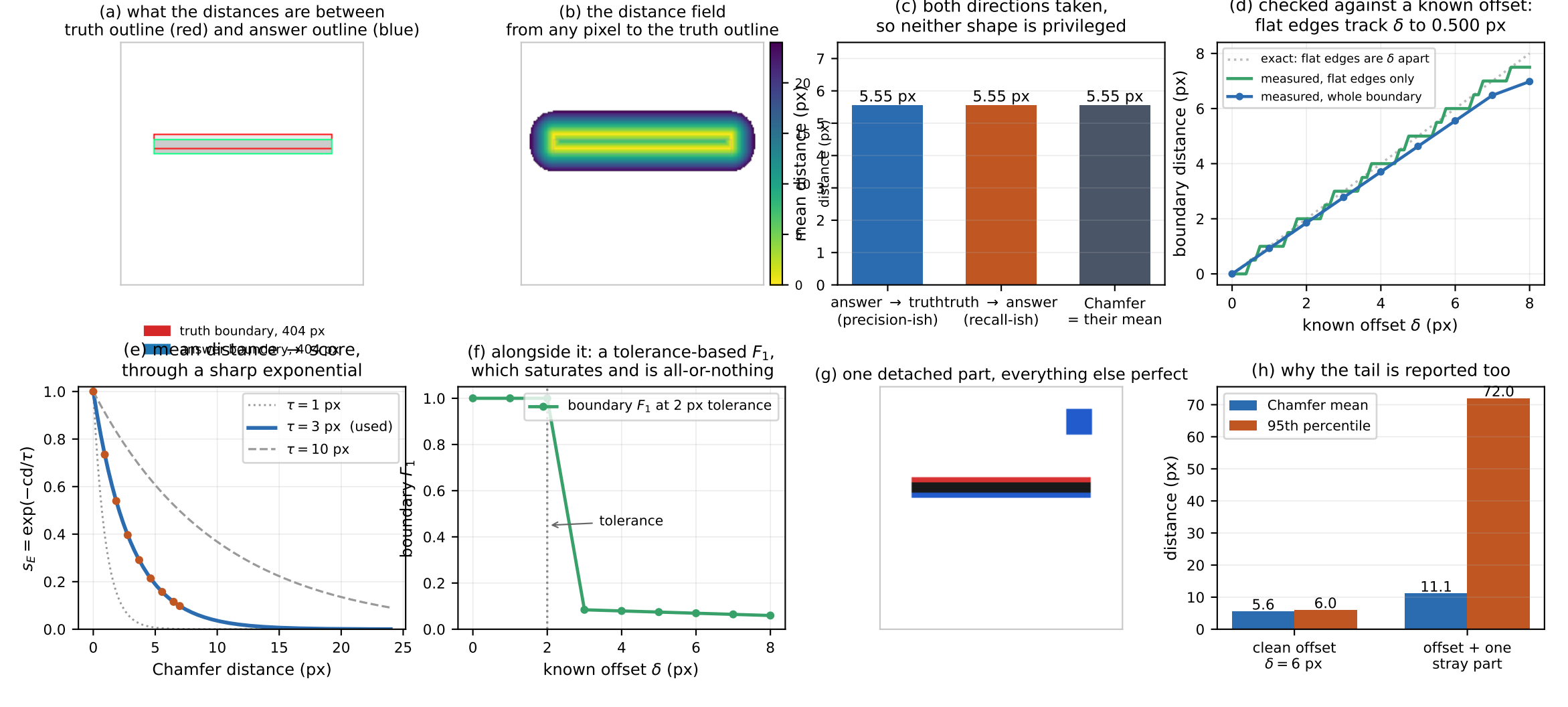
(a) what the distances are between
truth outline (red) and answer outline (blue)
(b) the distance field
from any pixel to the truth outline
distance (px)
mean distance (px)
(c) both directions taken,
so neither shape is privileged
5.55 px
5.55 px
5.55 px
answer → truth
(precision-ish)
truth → answer
(recall-ish)
Chamfer
= their mean
(d) checked against a known offset:
flat edges track δ to 0.500 px
exact: flat edges are δ apart
measured, flat edges only
measured, whole boundary
boundary distance (px)
known offset δ (px)
truth boundary, 404 px
answer boundary, 404 px
(e) mean distance → score,
through a sharp exponential
τ = 1 px
τ = 3 px (used)
τ = 10 px
$s_E = \exp(-cd/\tau)$
Chamfer distance (px)
(f) alongside it: a tolerance-based F_1,
which saturates and is all-or-nothing
boundary F_1 at 2 px tolerance
tolerance
boundary F_1
known offset δ (px)
(g) one detached part, everything else perfect
(h) why the tail is reported too
Chamfer mean
95th percentile
5.6
6.0
11.1
72.0
distance (px)
clean offset
δ = 6 px
offset + one
stray part